Question	Answer	Marks
1	**Defining the problem**	
	p is the independent variable and *B* is the dependent variable, or vary *p* and measure *B*.	1
	Keep the current/I (in the electromagnet) constant.	1
	Methods of data collection	
	Labelled diagram showing Hall probe correctly positioned (along *p*) and ruler correctly positioned and either Hall probe or rule supported.	1
	Correct circuit diagram to include d.c. power supply in series with coil and ammeter. Must be a workable circuit diagram to measure current through the coil.	1
	Measure *p* with ruler.	1
	Method to determine an accurate value of *p*. Examples include: Height of P above bench – height of electromagnet Height of P measured from ruler across the top of the electromagnet	1
	Method of analysis	
	Plot a graph of ln *B* against *p*.	1
	α = – gradient	1
	$k = \frac{e^{y\text{–intercept}}}{NI}$	1

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Question	Answer	Marks
	Additional detail including safety considerations	**6**
	1. Keep the number of turns/N constant. 2. Use large number of turns/current (to increase B). 3. Avoid overheating the coil/do not touch hot coil. 4. Use of variable resistor to keep ammeter reading constant. 5. Method to ensure that Hall probe is equidistant from the poles, e.g. determine centre of electromagnet and use of plumb line/ruler and spirit level/set square. 6. Adjust Hall probe until maximum reading obtained/perpendicular to field. 7. Repeat each experiment for the same value of p and reverse the current/Hall probe and average 8. $\ln B = -\alpha p + \ln kNI$ 9. Relationship is valid if the graph is a straight line. 10. Calibrate Hall probe using a known field.	

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